Hugo Armando Guillen Ramirez

Curriculum Vitae

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Objective

I am Ph.D. in Computer Science working on machine learning applied to RNA bioinformatics. My research interests are mostly machine learning, algorithm design, software development, optimization using bio-inspired strategies, bioinformatics, and graph theory applied to bioinformatics.

Education

2014–2018 **PhD in Computer Science**, *Center for Scientific Research and Higher Education at Ensenada (CICESE Research Center)*, Ensenada, Baja California, Mexico, Grade: 97.02/100.

2011–2014 **MSc in Computer Science**, *Center for Scientific Research and Higher Education at Ensenada (CICESE Research Center)*, Ensenada, Baja California, Mexico, Grade: 96.78/100.

2007–2011 **Bachelor in Computer Systems Engineering**, *Instituto Tecnológico de Tepic*, Nayarit, Mexico, Grade: 97/100.

Research experience

2019–present **Post-Doctoral Researcher**, *Project: "Identification and functional prediction of lncRNA elements."*, Advisor: Dr. Rory Johnson, University of Bern, Department for BioMedical Research, Bern, Switzerland.

2019–2019 **Post-Doctoral Researcher**, *Project: "Finding research opportunities in Artificial Intelligence for the public health of the Mexican population."*, Advisor: Dr. Salvador Villareal Reyes, CICESE Research Center/Artificial Intelligence Consortium, Mexico.

Doctoral thesis (2014–2018)

Title *Methods for non-coding RNA gene prediction* (translated from Spanish)

Description The main objective of this project is to develop a non-coding RNA (ncRNA) gene predictor based in homology search on known sequences supported by machine learning, meta-heuristics, and state-of-the-art biological information.

Supervisor Dr. Israel Marck Martínez Pérez, CICESE Research Center, Mexico

Master thesis (2011-2014)

Title *Design of a Tissue P-System and a molecular algorithm to solve the MAX-CLIQUE problem* (translated from Spanish)

Description This theoretical work presents a tissue P-System that solves the MAX-CLIQUE problem in polynomial time, and also a constructive DNA algorithm inspired by this system.

Supervisor Dr. Israel Marck Martínez Pérez, CICESE Research Center, Mexico

Languages

Spanish Native

English Proficient

Indexed publications

Journal Bosch-Guiteras, N., Uroda, T., **Guillen-Ramirez, H.A.**, Riedo, R., Gazdhar, A., Esposito, R., Pulido-Quetglas, C., Zimmer, Y., Medová, M. & Johnson, R. (2021). Enhancing CRISPR deletion via pharmacological delay of DNA-PKcs. Genome Research. https://doi.org/10.1101/gr.265736.120

Journal Aguilera-Mendoza, L., Marrero-Ponce, Y., García-Jacas, C. R., Chavez, E., Beltran, J. A., **Guillen-Ramirez, H. A.**, & Brizuela, C. A. (2020). Automatic construction of molecular similarity networks for visual graph mining in chemical space of bioactive peptides: an unsupervised learning approach. Scientific reports, 10(1), 1-23. https://doi.org/10.1038/s41598-020-75029-1

Journal Calvillo-Vázquez, J. G., **Guillén-Ramírez, H. A.**, DiazDuarte-Rodríguez, M., Licea-Claverie, A., & Méndez, E. R. (2019). Particle size distribution from extinction and absorption data of metallic nanoparticles. Applied optics, 58(36), 9955-9966. https://doi.org/10.1364/AO.58.009955

Journal Aguilera-Mendoza, L., Marrero-Ponce, Y., Beltran, J. A., Tellez Ibarra, R., **Guillen-Ramirez, H. A.**, & Brizuela, C. A. (2019). "Graph-based data integration from bioactive peptide databases of pharmaceutical interest: toward an organized collection enabling visual network analysis." Bioinformatics. https://doi.org/10.1093/bioinformatics/btz260

Journal **Guillen-Ramirez, H. A.**, & Martínez-Pérez, I. M. (2018). "Classification of riboswitch sequences using k-mer frequencies". Biosystems, 174, 63-76. https://doi.org/10.1016/j.biosystems.2018.09.001

Conference Ibarra-Hernández, R. F., Bertin, N., Alonso-Arévalo, M. A., & **Guillén-Ramírez, H. A.** (2018). "A benchmark of heart sound classification systems based on sparse decompositions". In Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series (Vol. 10975). https://doi.org/10.1117/12.2506758

Conference **Guillen-Ramirez, H. A.**, Colbes, J., Brizuela, C. A., & Martinez-Perez, I. M. (2017). "Accurate classification of immunomodulatory RNA sequences". In 2017 International Joint Conference on Neural Networks (IJCNN) (pp. 236-241). IEEE. https://doi.org/10.1109/IJCNN.2017.7965860

Conference Brizuela, C. A., Luna-Taylor, J. E., Martinez-Perez, I., **Guillen, H. A.**, Rodriguez, D. O., & Beltran-Verdugo, A. (2013). "Improving an evolutionary multi-objective algorithm for the biclustering of gene expression data". In 2013 IEEE Congress on Evolutionary Computation (pp. 221-228). IEEE. https://doi.org/10.1109/CEC.2013.6557574

Selected presentations

2020, Oct Seminar **Discrete mathematics and the design of CRISPR-Cas9 genome editing libraries** at Instituto Tecnológico de Tepic, Mexico.

2017, May Oral presentation **Accurate Classification of Immunomodulatory RNA Sequences.** International Joint Conference on Neural Networks 2017, Anchorage, Alaska, USA.

2017, Mar Seminar **Score functions for discriminant k-mers and their application in ncRNA classification** (translated from Spanish) at CICESE Research Center, Mexico.

2016, Oct Seminar **Methods for ncRNA gene prediction: RNA secondary structure alternative graph representations** (translated from Spanish) in the XII National Meeting on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.

2016, Oct Hands-on **Bio-computing: RNA, BioPython, and Graphs** (translated from Spanish) in the Computer Science Fall School at CICESE Research Center, Mexico.

Awards

2021–2023 Marie Skłodowska-Curie Individual Fellowship (European Commission; Host institution: University College Dublin, Ireland).

2021–2024 "Candidate to National Researcher" member of the Mexican National System of Researchers (SNI) (Mexico).

2014–2018 CONACYT National Scholarship for doctoral studies (Mexico).

2011–2013 CONACYT National Scholarship for master studies (Mexico).

2011 Award for academic achievement (best grade in class 2006-2011). Instituto Tecnológico de Tepic (Mexico).

Experience

Teaching

2019–2019 **Computational Biology** course (Genome indexing algorithms) at CICESE Research Center, Mexico [Graduate level].

2017–2019 **Artificial Intelligence**, **Advanced Mathematics**, **Evolutionary Computing**, and **Assembly Language** courses at Universidad Autónoma de Baja California, Ensenada, Mexico [Undergraduate level].

2017–2018 **Mathematics II, IV, and V** (in English), **Intermediate Algebra I and II** (in English) and **Mathematical reasoning V and VI** (in Spanish) at Xochicalco High School, Ensenada, Mexico [High school level].

2012–2012 Teacher assistant of the **Discrete Mathematics** course at CICESE Research Center. [Graduate level].

Attendant

2021 Data Management Plan (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2020 Enrichment Analysis, (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2020 Advance statistics: statistical modeling, (Swiss Institute of Bioinformatics), Lausanne, Switzerland.

2016 XVIII Fall School on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Michoacán, Mexico.

2016 NGS Data Analysis and Informatics Conference, San Diego, USA.

2015 Applications of Artificial Organic Networks (National Meeting on Computer Science 2015), Baja California, Mexico.

Developer

2019–present University of Bern, Bern, Switzerland. In my current postdoc project I have been developing tools for feature engineering, automatic machine learning, network visualization, statistical testing on genomic regions, and pipelines for designing and processing CRISPR deletion libraries.

2013–2018 CICESE Research Center, Ensenada, Mexico. During my doctoral work I developed and implemented machine learning techniques, meta-heuristics, and visualisation tools for classification problems involving genomic sequences and RNA secondary structures.

2016 Developer of REFRACTIVEINDEX.INFO-SQLITE, a Python 3 open source package to locally process the refractiveindex.info database. `http://github.com/HugoGuillen/refractiveindex.info-sqlite` (13 4 ★).

2012 MXGlobal Solutions, Nayarit, Mexico. I developed C# and SQL software.

2010-2011 Instituto Tecnológico de Tepic, Nayarit, Mexico. I developed a web system in Java/JSP.